

Week 3

$$1.94 \text{ slug/ft}^3$$

$$\rho = 1.94 \frac{\text{slug}}{\text{ft}^3} \rightarrow \text{SI}$$

$$\rho = 1.94 \frac{\text{slug}}{\text{ft}^3} \left(\frac{14.59 \text{ kg}}{1 \text{ slug}} \right) \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3$$

$$\rho = 1.94 (14.59) \left(\frac{1}{0.3048} \right)^3$$

$$= 999.8 \text{ kg/m}^3$$

$$= 1000 \text{ kg/m}^3$$

$$= 1.00 \times 10^3 \text{ kg/m}^3$$

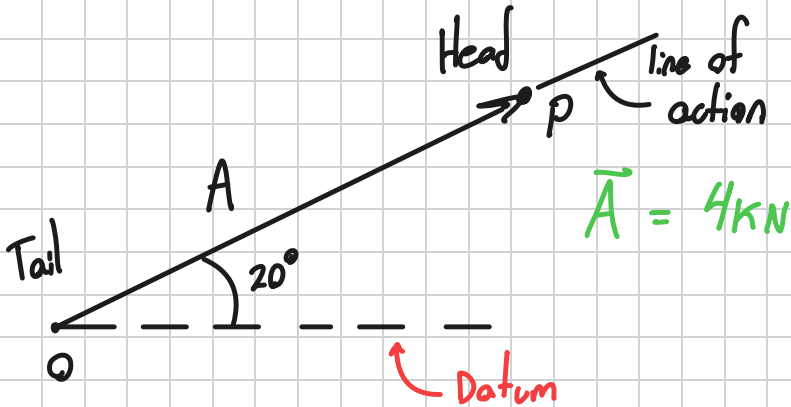
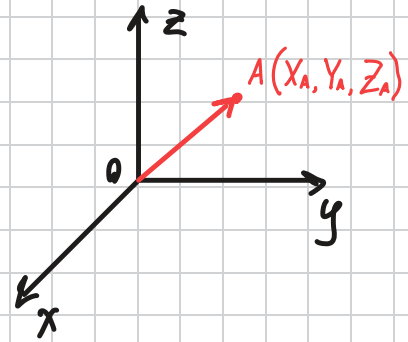
$$= 1.00 \cdot 10^6 \text{ g/m}^3$$

$$= 1.00 \text{ Mg/m}^3$$

Vectors

=> Magnitude
=> direction

- Force vectors
- moment vectors
- position vectors
- Displacement vectors



- point of Application

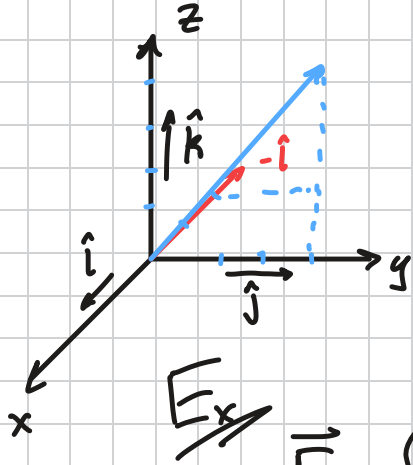
tension

compression

Unit vector

$$\textcircled{U_A} = \frac{A}{A} \begin{matrix} \nearrow k_N \\ \searrow k_N \end{matrix} \Rightarrow \vec{A} = |\vec{A}| \vec{u}_A$$

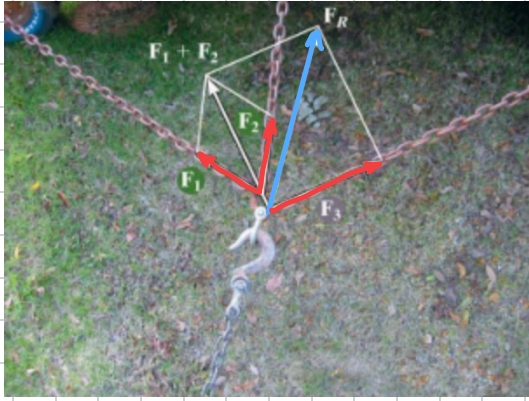
dimensionless



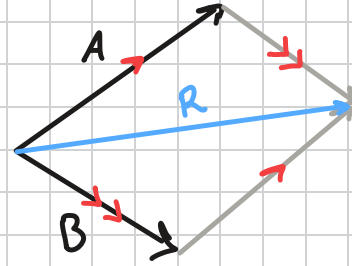
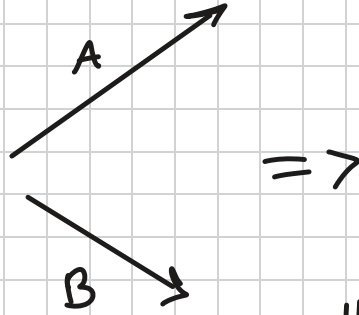
Cartesian unit Vectors

~~E_x~~

$$\vec{F} = \{-2\hat{i} + 3\hat{j} + 4\hat{k}\}$$

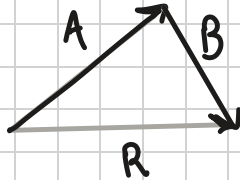
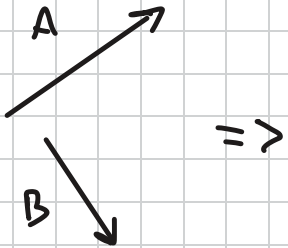


Vector addition

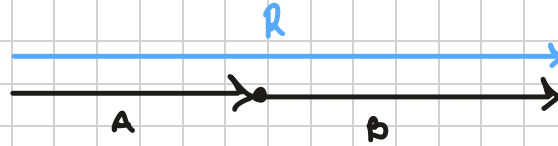


Parallel Law

triangle rule



Special case



Cosine Law

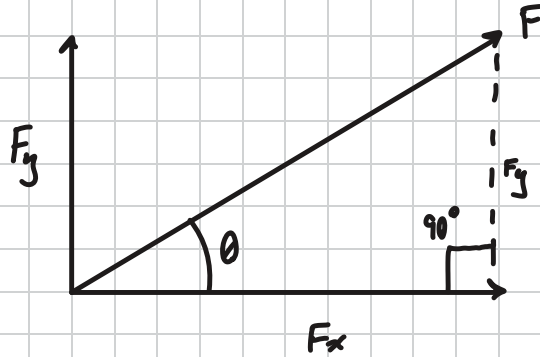
$$C = \sqrt{A^2 + B^2 - 2AB \cos C}$$

Sin Law

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

Vectors $\rightarrow$ 2D Analysis

Cartesian Vectors



$$F = F_x + F_y$$

$$\Rightarrow F_x = F \cos \theta$$

$$\Rightarrow F_y = F \sin \theta$$

$$\frac{F}{\sin \theta} = \frac{F_y}{\sin \theta} \rightarrow \therefore F_y = F \sin \theta$$

$$\frac{F}{\sin \theta} = \frac{F_x}{\sin(90^\circ - \theta)}$$

$$\vec{F} = F_x \vec{i} + F_y \vec{j}$$

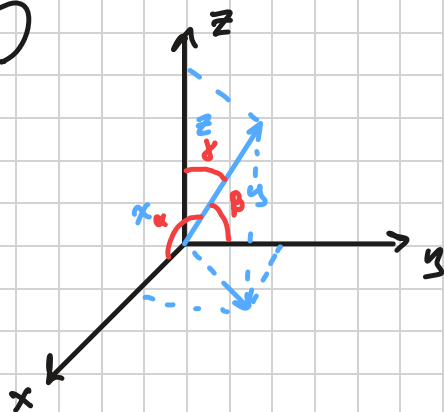
$$\Rightarrow F = \sqrt{F_x^2 + F_y^2}$$

$$\Rightarrow \tan \theta = \frac{F_y}{F_x}$$

$$\Rightarrow F_x = F \cos \theta$$

$$\Rightarrow F_y = F \sin \theta$$

3D



$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z$$